

LAB Activity-1

⇒ GRAPH OF CS DEPARTMENT



TRANSITION TABLE:

Gate 1 → Lab 1 → Room → Mess Hall
→ Miss & Mrs office → Girls Common Room →
DR Akhtar office → Sir Khalid office →
CS dept. office → Rooms.

Initial = Gate 1

Goal state = Room No 1

PROBLEM FORMULATION:

Initial state → Gate 1

Goal state → Room No. 1

States → Gate 1, Lab 1, Rooms, Hall
Teachers offices, dept offices.

Actions → Move from one connected
location to another.

Transition Model → A person can move
between two locations
if they are directly connected

Goal Test → Check whether the current
location is Room No. 1

Initial state $\rightarrow$ Gate 1

Goal state $\rightarrow$ Room No 1

Objective $\rightarrow$ Find the shortest path from Gate 1 to Room No 1 using BFS.

BFS TABLE:

0	Gate 1	Lab 1
1	Lab 1	Room
2	Room	Hall
3	Hall	Miss Salwa office
4	Miss Salwa office	corner
5	Common Room	Dr Akhtar office
6	Dr Akhtar Office	Sir Khalid office
7	Sir kashid office	cs dept.
9	dept office	Room 1
16	Room No 1	Goal Reached.

SHORTEST PATH

